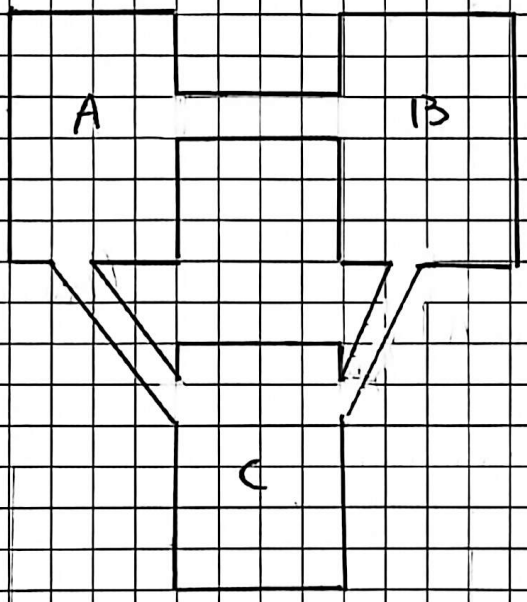
LECTURE 3

$$S = -\sum_i P(i) \log P(i) \qquad \sum_i P(i) = 1$$

$$\sum_i P(i) E(i) = \langle E \rangle$$

1. ☆ Zeroth law of thermodynamics:



A and B in equilibrium $\Leftrightarrow T_A = T_B$

Meaning of temperature:
Tells you the direction of the heat flow

2. $T_B > T_A$

1st law: $dE_A + dE_B = 0$ (1)

2nd law: $dS_A + dS_B > 0 \Leftrightarrow dS > 0$ (2)

$$T = \frac{dE}{dS} \Rightarrow \begin{cases} dE_A = T_A \, dS_A \\ dE_B = T_B \, dS_B \end{cases}$$

Sub (1):

$$\Rightarrow T_A dS_A + T_B dS_B = 0$$

$$\Rightarrow dS_B = -\frac{T_A}{T_B} dS_A$$

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Sub (2)

$$\Rightarrow dS_A - \frac{T_A}{T_B} dS_A > 0$$

$$\Rightarrow T_B dS_A - T_A dS_A > 0$$

$$\Rightarrow (T_B - T_A) dS_A > 0$$

We just assume that $T_B - T_A > 0$

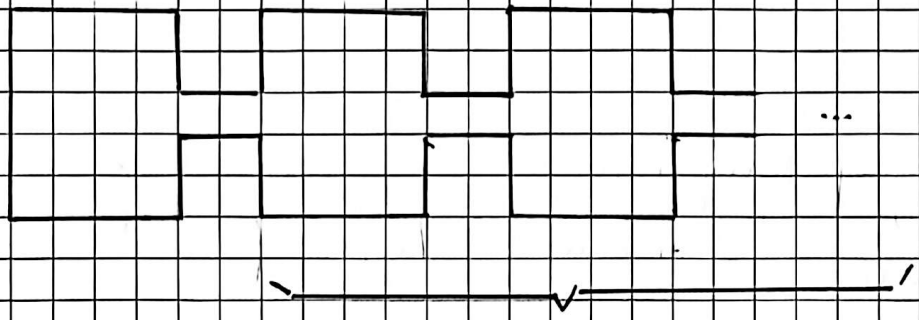
$$\therefore dS_A > 0$$

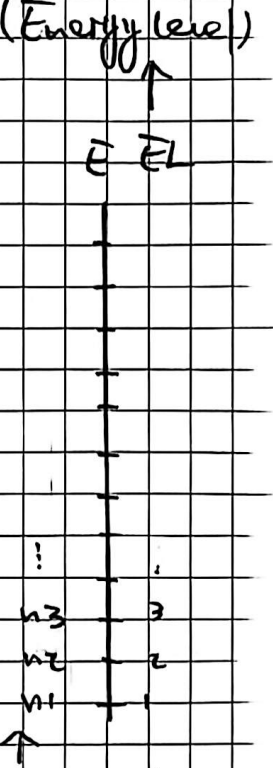
$$\Rightarrow T_A dS_A > 0$$

$$\Rightarrow d\bar{E}_A > 0$$

Energy has flew from B to A

3. Heat bath & Occupation Number





Constraints:

$$n_1 + n_2 + \cdots = \sum_{\bar{i}} n_{\bar{i}} = N$$

$$n_1 \bar{E}_1 + n_2 \bar{E}_2 + \cdots = \sum_{\bar{i}} n_{\bar{i}} \bar{E}_{\bar{i}} = N\bar{E}$$

$\bar{E}$: Average $\bar{E}$ per subsyste

$n_{\bar{i}} = n_1, n_2, \cdots$

$$P_{\bar{i}} = \frac{n_{\bar{i}}}{N} \quad \Leftrightarrow \quad n_{\bar{i}} = P_{\bar{i}} N$$

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$$\sum_i n_i = N$$

$$\Rightarrow \sum_i n_i = \sum_i p(i) N = N \sum_i p_i$$

$$\Rightarrow \sum_i p_i = 1$$

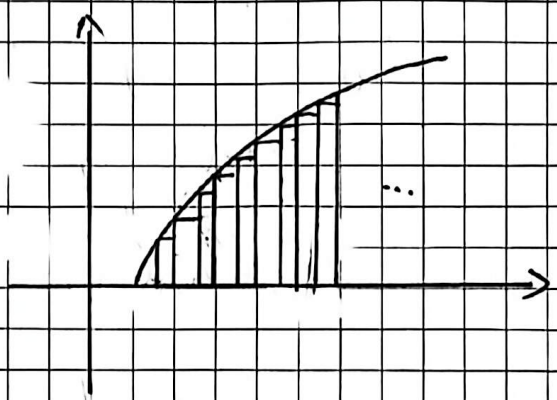
$$\sum_i p_i E_i = \bar{E}$$

Given a set of occupation numbers, n_i, how many ways are there of redistributing our systems to correspond to that number ?

$$\text{number of arrangements} = \frac{N!}{\prod_i n_i!} \qquad (0! = 1)$$

4. Sterling's approximation

$$N! = N^N e^{-N} \longrightarrow \text{converge very slow}$$



$$\log N! \qquad \log \to \ln$$

$$= \log 1 + \log 2 + \cdots + \log N$$

$$= \sum_{x=1}^{N} \log x$$

$$= \int_1^N dx \log x$$

$$\frac{d}{dx}(x \log x - x) = \log x$$

$$= \left[x \log x - x \right]_1^N \qquad (\text{when } N \to +\infty,\ 1 \text{ is negligible})$$

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$$\approx N\log N - N$$

$$\therefore \log N! = N\log N - N$$

$$\Rightarrow N! = e^{N\log N} e^{-N}$$

$$\Rightarrow N! = N^N e^{-N}$$

5. Applying the approximation:

$$\text{Number of arrangements } (n_1, n_2, \ldots) = \frac{N!}{\prod_i n_i!}$$

$$C = \frac{N^N e^{-N}}{n_1^{n_1} n_2^{n_2} n_3^{n_3} e^{-(n_1+n_2+n_3)}} = \frac{N^N}{n_1^{n_1} n_2^{n_2} n_3^{n_3}}$$

$$\Rightarrow \log C = N\log N - n_1\log n_1 - n_2\log n_2 - n_3\log n_3$$

$$\Rightarrow \log C = N\log N - \sum_i n_i \log n_i$$

$$= N\log N - \sum_i N p_i \log(N p_i)$$

$$= N\log N - \sum_i N p_i (\log p_i + \log N)$$

$$= -N\sum_i p_i \log p_i \qquad (\log p_i < 0)$$

$$= -NS$$

$$\uparrow$$

Entropy

$$C = e^{-NS} \qquad S\downarrow \Rightarrow C\uparrow$$

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finding a probability distribution p_i

letting $-\sum_i p_i \log p_i$ under the

constraints of $\sum_i p_i = 1$ and $\sum_i p_i E_i = \bar{E}$

(Maximize something with constraints)

6. Lagrange Multipliers

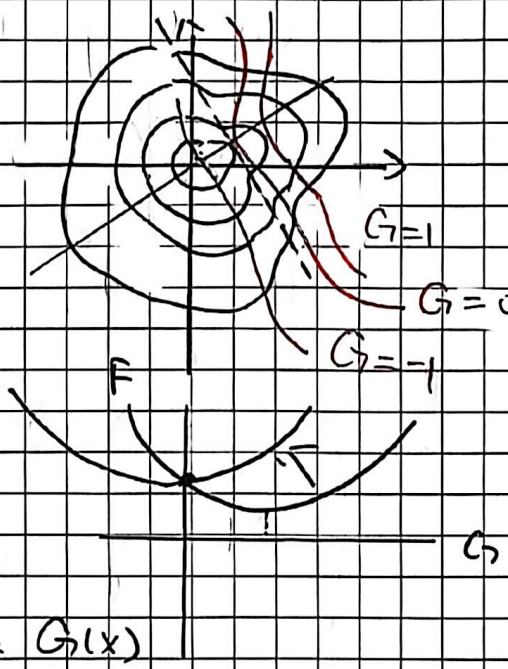
$F(x_i)$

$\frac{\partial F}{\partial x_1} = 0$

$\frac{\partial F}{\partial x_2} = 0$

$\frac{\partial F}{\partial x_3} = 0$

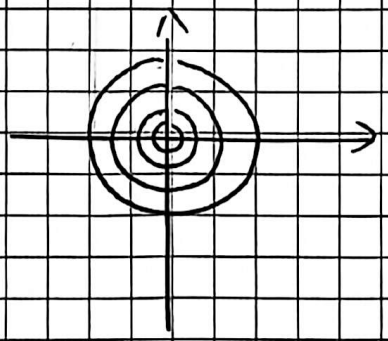
$\vdots$



Constraints Equation: $G(x)$

Where is F be maximum along the constraint curve?

$F' = F + \lambda G$



e.g. $\frac{x^2+y^2}{2}\Big|_{\min}$ $x+y=1$

$F = \frac{x^2+y^2}{2}$, $G = x+y-1$

$F' = \frac{x^2+y^2}{2} + \lambda(x+y-1)$

$\begin{cases} \frac{\partial F'}{\partial x} = x+\lambda = 0 \Rightarrow x = -\lambda \\ \frac{\partial F'}{\partial y} = y+\lambda = 0 \Rightarrow y = -\lambda \end{cases}$

$G = 0 \Rightarrow x+y = 1$

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$\Rightarrow -2\lambda = 1$

$\Rightarrow \lambda = -\frac{1}{2}$

e.g. $F = \frac{x^2+y^2-z^2}{2}$

$G_1 : (x+y) = 1$

$G_2 : (z+x) = 1$

$F' = \frac{x^2+y^2-z^2}{2} + \lambda_1(x+y-1) + \lambda_2(z+x-1)$

$$\begin{cases} \frac{\partial F'}{\partial x} = x + \lambda_1 + \lambda_2 = 0 \\ \frac{\partial F'}{\partial y} = y + \lambda_1 = 0 \\ \frac{\partial F'}{\partial z} = -z + \lambda_2 = 0 \end{cases}$$

$x+y=1$

$x+z=1$

$x+y+2\lambda_1+\lambda_2 = 0$

$\Rightarrow 2\lambda_1 + \lambda_2 = -1 \quad (1)$

$x+z+\lambda_1 = 0$

$\Rightarrow \lambda_1 = -1 \quad (2)$

$$\begin{cases} \lambda_1 = -1 \\ \lambda_2 = 1 \end{cases}$$

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General Problem:

$$\begin{cases} F(p_1, p_2, \dots, p_n) & \text{(e.g. Entropy)} \\ G_1 = \sum_i p_i - 1 \\ G_2 = \sum_i p_i E_i - E \end{cases}$$

Find max/min F

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